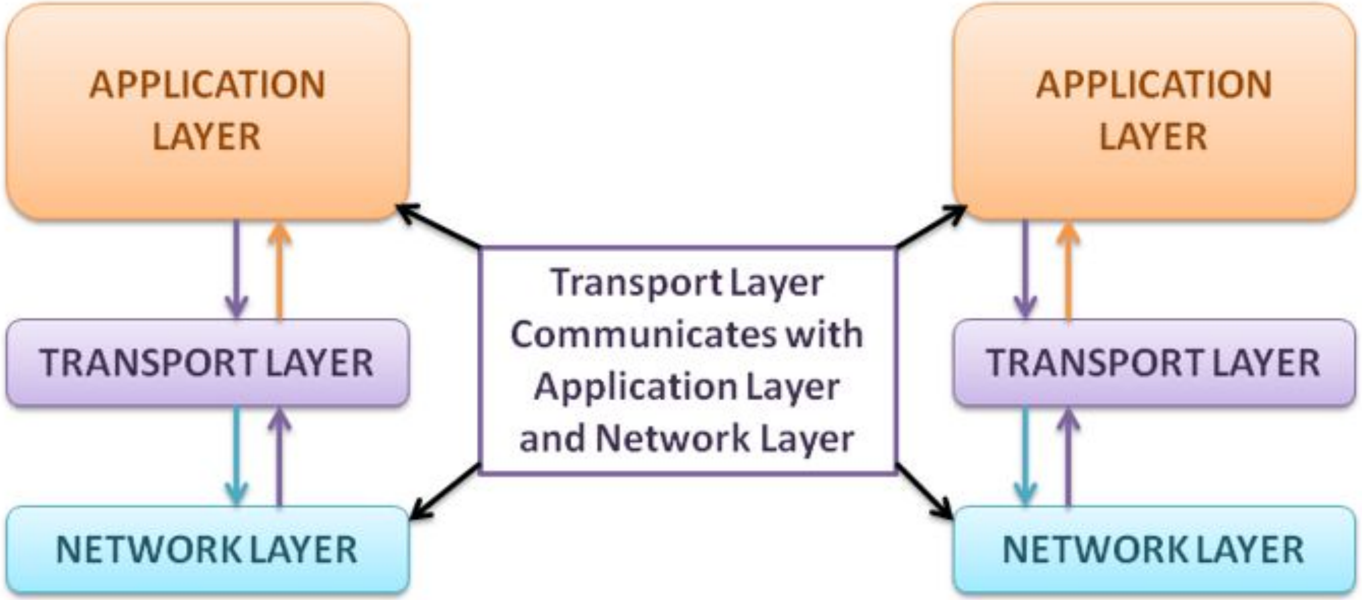
# Transport Layer & Application Layer

- Transport Layer:
  - Transport services, Elements of Transport Protocol, UDP, TCP, Socket Programming (TCP & UDP), TCP Flow control, TCP Congestion Control.
- Application Layer:
  - Application Layer Protocols, Application layer protocols interaction with end-user applications, Presentation and Session layers. Well-Known Application Protocols and Services, Domain Name System (DNS), Hypertext Transfer Protocol (HTTP).

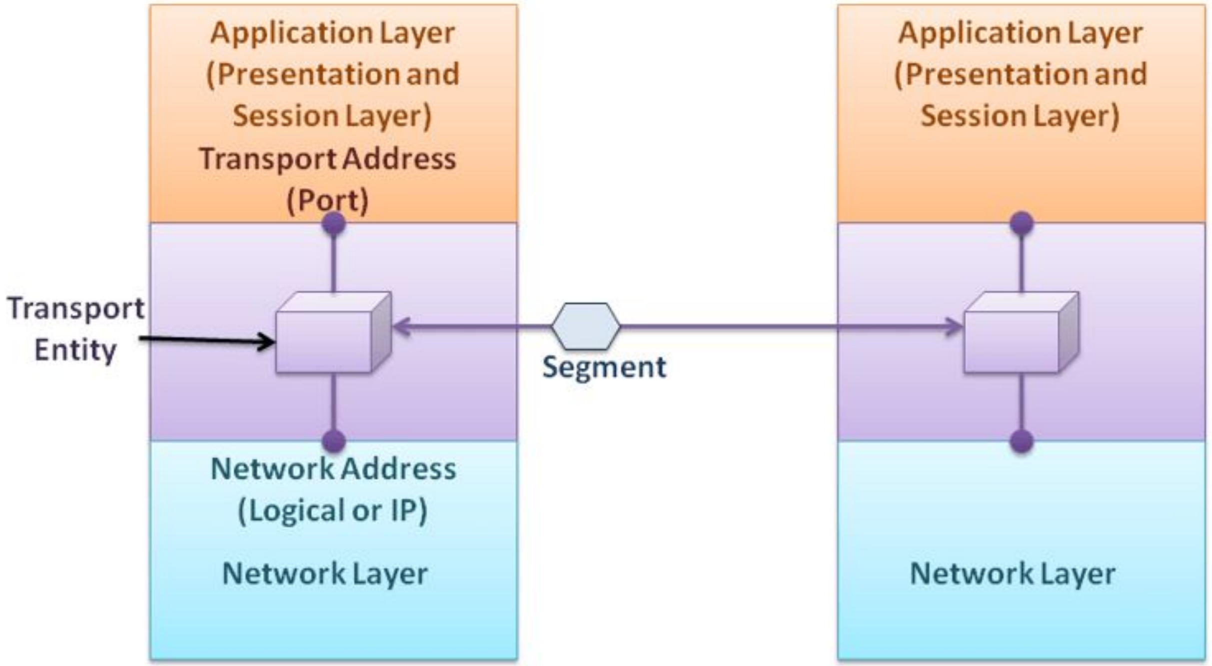
# Transport Layer

- The transport layer receives the data generated at the application layer.
- Its main objective is process-to-process delivery.
- The logical communication between the sender’s applications and the receiver’s applications is the responsibility of the transport layer.



# Transport Layer services

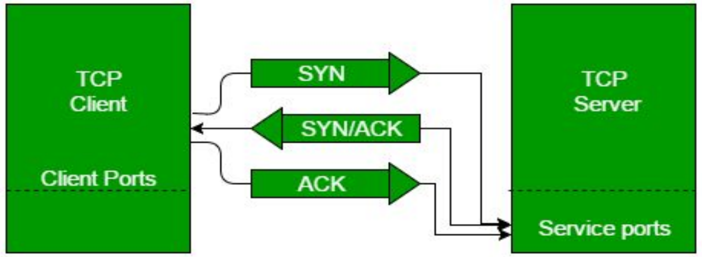
- Services Provided to the Upper Layers
  - The transport layer communicates with the network layer and uses the services provided by the network layer to achieve efficient, reliable, and cost-effective data transmission service.
  - This work is done by the transport entity, which is software or hardware.
  - Transport layer services are mostly similar to network layer services. But the network layer mostly runs on the routers, and the transport layer runs entirely on the end-users machines.
  - Here, the network layer and the transport layer are different because users have no real control over the network layer, so they cannot solve poor service by adding better-performing routers or doing more error handling.



# Elements of Transport Protocol

- End-to-end Connection between Hosts
  - The transport layer is also responsible for creating the end-to-end Connection between hosts, for which it mainly uses TCP and UDP
- Error Control and Flow Control
  - Error control is ensuring that the data is delivered with the desired level of reliability, usually that all of the data is delivered without any errors.
  - Flow control is keeping a fast transmitter from overrunning a slow receiver.
- Multiplexing and Demultiplexing
  - Multiplexing(many to one) is when data is acquired from several processes from the sender and merged into one packet along with headers and sent as a single packet.
  - Similarly, Demultiplexing(one to many) is required at the receiver side when the message is distributed into different processes.
- Addressing
  - Transport Layer deals with addressing or labelling a frame. It also differentiates between a connection and a transaction.

- Connection Establishment
  - When two devices in a network wants to establish a connection using TCP, it is done through 3-Way Handshake Process
  - The first computer connects to the second computer by sending a SYN packet to a specified port number.
  - If the second computer is listening, it will respond with a SYN/ACK.
  - When the first computer receives the SYN/ACK, it replies with an ACK packet.
  - After this, the two devices can communicate normally.



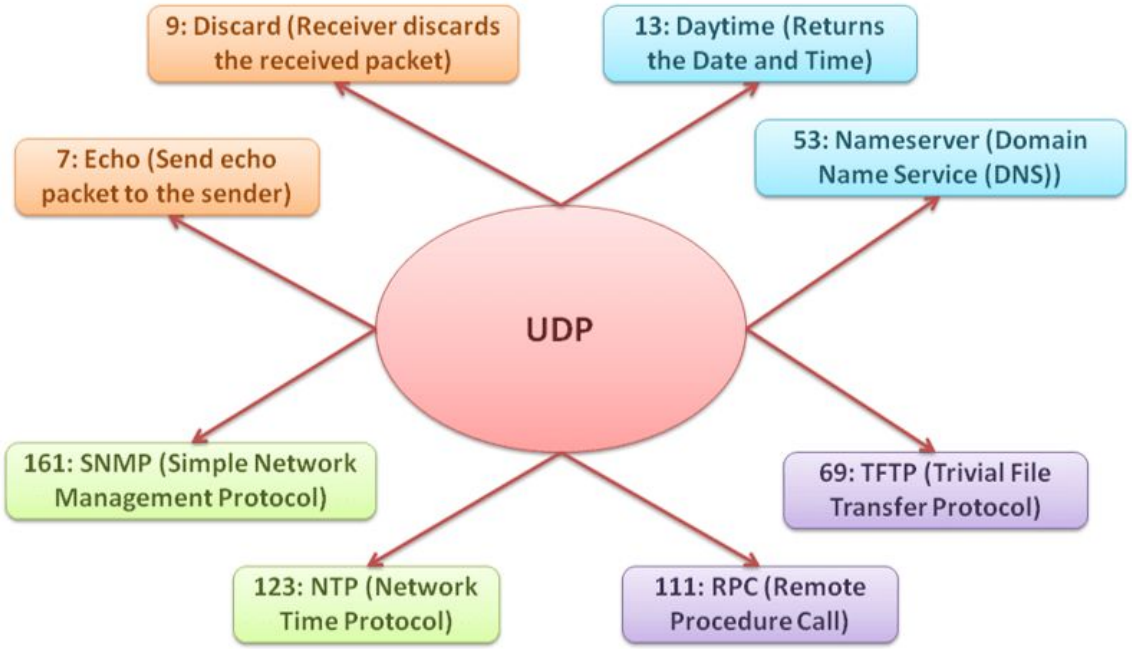

- Connection Termination
  - In a TCP connection, we have two types of termination mechanisms:
    - In the Graceful connection release, the connection is open until both parties have closed their sides of the connection.
    - In an Abrupt connection release, either one TCP entity is forced to close the connection or one user closes both directions of data transfer.
- Crash Recovery
  - It may be possible that hosts or routers on the network are subject to crashes, or the connection is established for a long time such as large software or media downloads.

# User Datagram Protocol (UDP)

- It provides a connectionless service, or you can say it is a connectionless protocol.
- Basically, it just sends packets from sender to receiver, which means that the path between sender and receiver is not established.
- UDP does not guarantee packet delivery.
- The transport layer receives data from the application and creates segments by adding UDP headers to the data.
- The UDP protocol transmits segments from sender to receiver before establishing a connection.
- In general, UDP is preferable if data has to be sent quickly, regardless of packet loss.
- Because UDP is a connectionless protocol, it is also known as a stateless protocol, which means that it does not track the state of the communication session.
- Applications that tolerate less data loss can use the UDP protocol. For example, audio and video conferencing tolerate some data loss, but communication is faster.

# Ports of UDP

- UDP has port numbers that are used when the messages are transmitted from sender to receiver.
- Generally, when UDP is used for communication, there is less interaction between sender and receiver, which increases the transmit data rate.



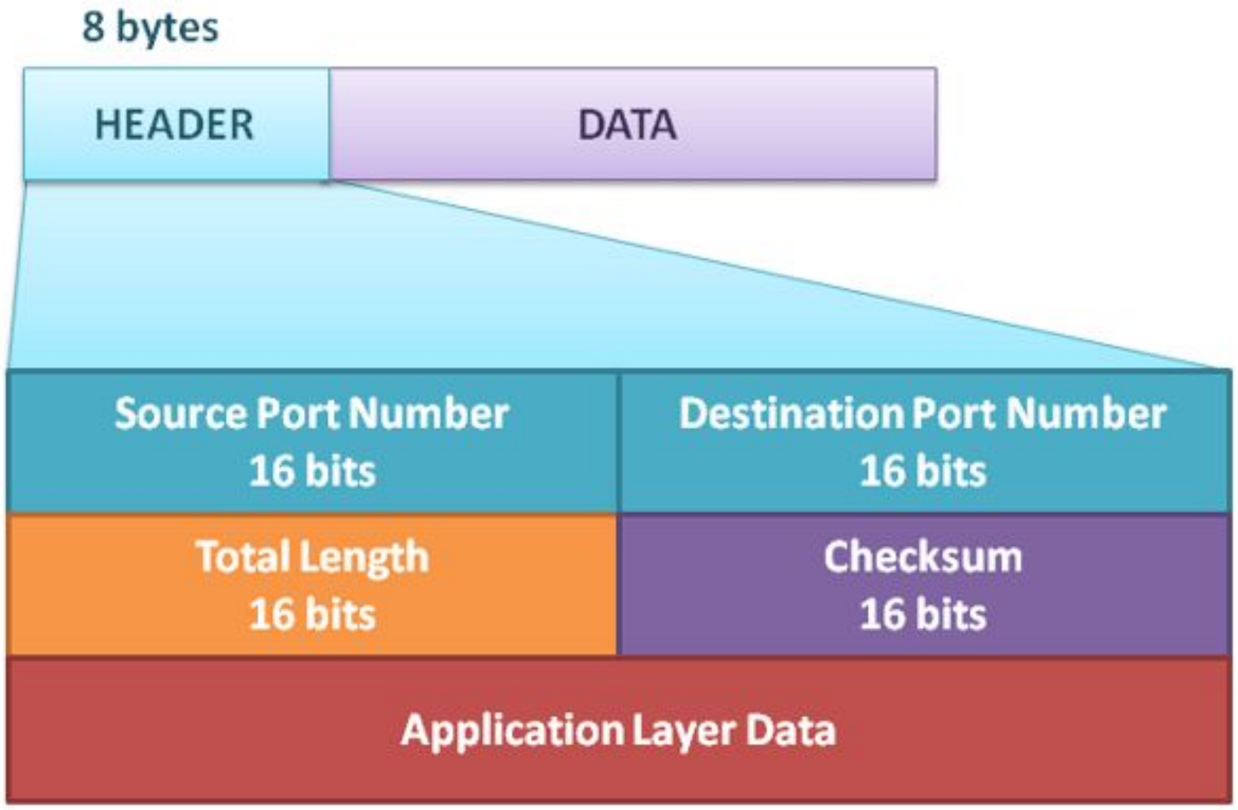
# UDP Header

- Source Port Number: The size of the source port number is 16 bits, which means that it has a range of 0 to 65535. It runs on the process of the source machine. If the source is a client and wants a request to the server, it uses the ephemeral port number, and the destination port number is set to the well-known port number.

- Destination Port Number: The size of the destination port number is 16 bits. If the destination is the server, the port number is the well-known port address, and if it is the client, the port number is ephemeral.

- Total Length: The Total Length field is used to define the length of the user datagram or packet, headers, and data. Normally, a UDP datagram is encapsulated in the IP header at the network layer, so in most cases, it is not necessary to have a length field in a UDP datagram because IP packets have a length field that defines the total length.

- Checksum: The checksum is an error detection method used to detect errors during communication across the entire UDP datagram.



# Uses of UDP

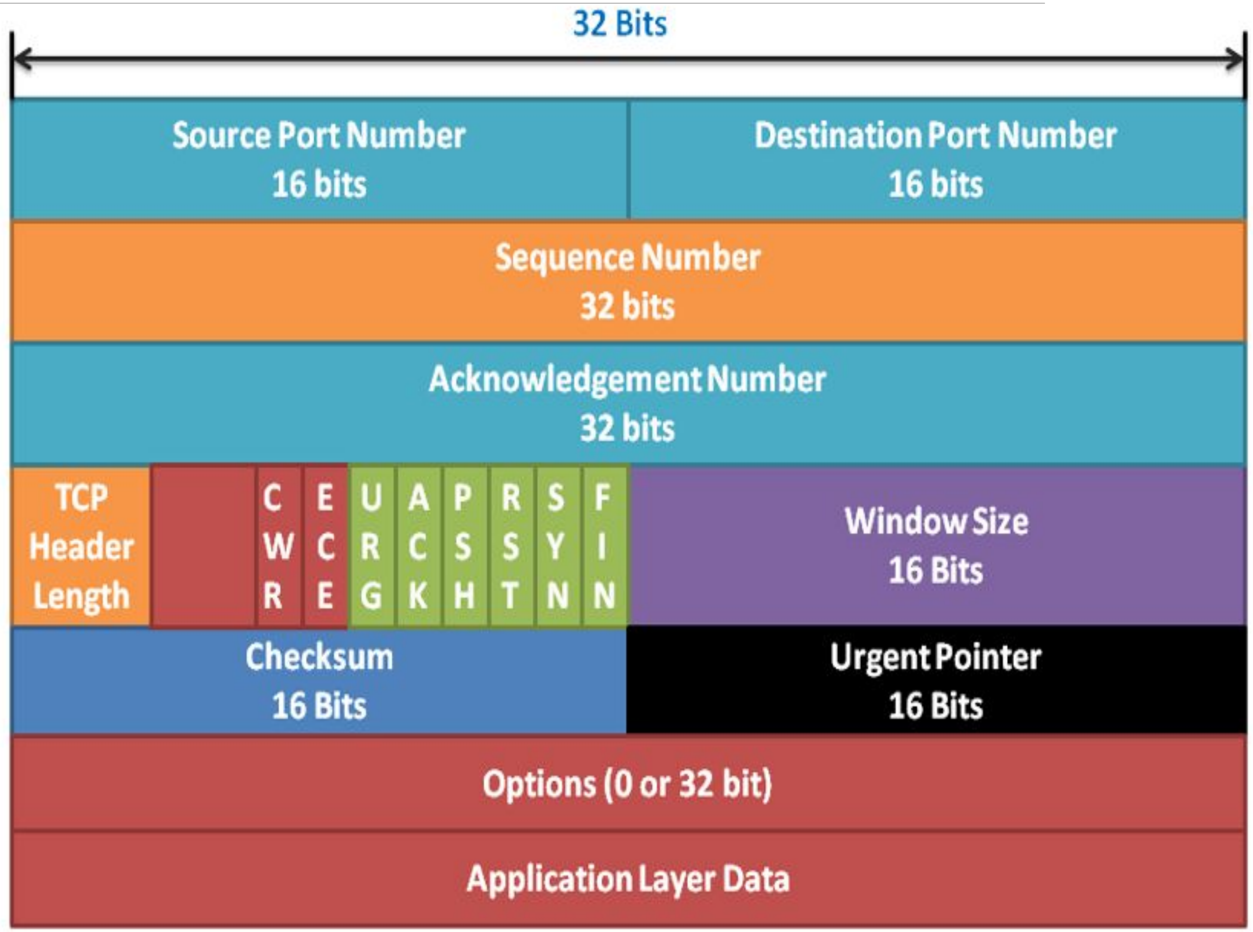
- If flow and error control are not required, and the process requires simple request-response communication, the UDP protocol is suitable for communication.
  - For example, DHCP and DNS in which the end device sends the request but it may not get a reply back.
- Generally, UDP does not have flow and error control but is used for processes that have their own flow and error control mechanisms.
  - For example, TFTP and SNMP have their own flow and error control mechanisms, so UDP is used for faster transmission.
- UDP is good for live video and multimedia applications as these applications can tolerate some data loss but require fast communication.
  - For example, video conferencing, live streaming of video, and Voice over IP (VoIP).
- UDP is also used in RIP (Routing Information Protocol), one of the route updating protocols.

# Transmission Control Protocol (TCP)

- TCP is one of the protocols of the transport layer. It is a reliable and connection-oriented protocol. Originally, it is designed to provide reliable communication over an unreliable network.
- When TCP segments travel through the Internetwork, they pass through multiple paths and different types of networks. They dynamically optimize the properties of the internetwork, which prevents TCP segment failures.
- TCP has the functionality to establish a session between sender and receiver. It provides reliable communication, same-order delivery, and flow control.
- The transport layer divides the data into TCP segments so that they can fit into IP datagrams and Ethernet frames. The IP protocol uses a best-effort mechanism, so there is no guarantee that datagrams will be delivered properly; that's all TCP determines.
- TCP is reliable and slower than UDP because it has to manage and track segment information.
- The TCP protocol provides the reliability that is needed for most applications. It is also used in the segmentation and reassembly of data.

# TCP Segment Header

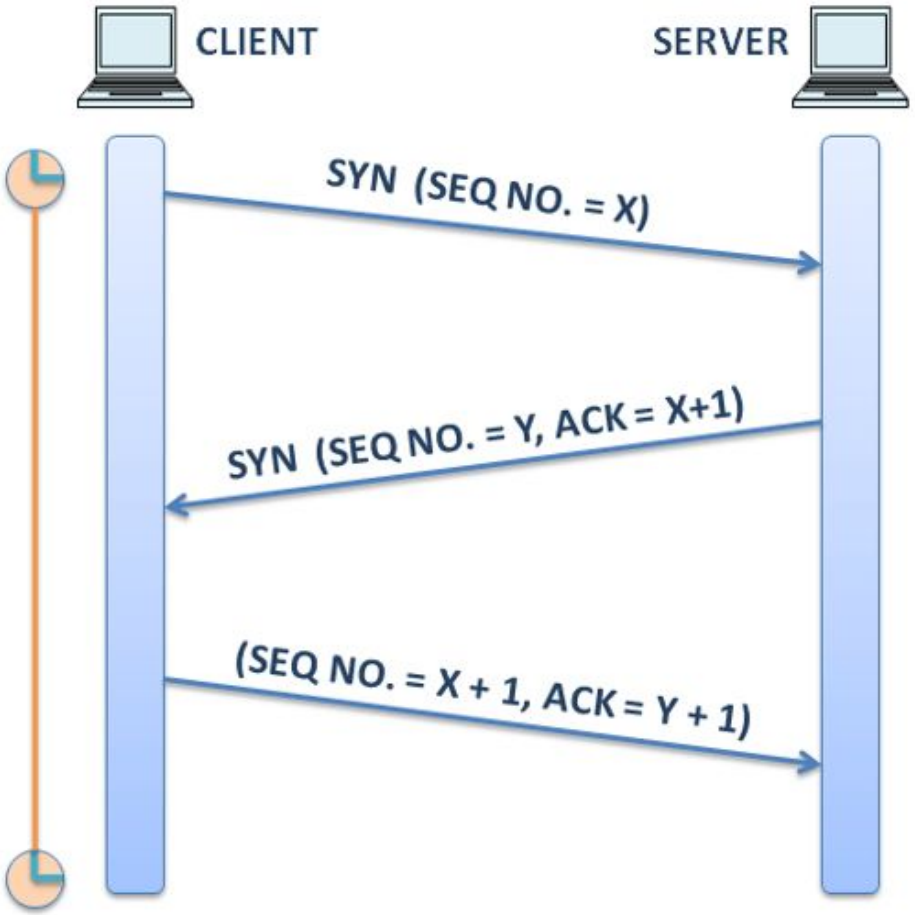
- TCP is a stateful protocol which means that it manages and keeps track of the state information of each segment and session.
- TCP records that when it sends a segment and which segment was accepted.
- All details of the data are stored in the TCP header, which is 20 bytes.
- When the transport layer receives data from the application layer, it adds a TCP header to the data to form a TCP segment.



- Source and destination ports: The source and destination ports are 16 bits used to identify the source and destination application, respectively.
- Sequence Number: The sequence number is a 32-bit field used on the receiver side to reassemble the data.
- Acknowledgment Number: The acknowledgment number is a 32-bit field sent by the receiver to the sender to say that the data has been received.
- Header Length: The Header Length field is of 4 bits which indicates the length of the TCP segment header.
- CWR (Congestion Window Reduced) and ECE (ECN-Echo): Both fields are 1-bit flags used when ECN is used.
- The ECE flag is set if the receiver wants to tell the sender to slow down the communication.
- The CWR flag is set by the sender to let the receiver know that it has slowed down the rate of data transmission.
- URG (Urgent Pointer): When urgent data is detected during communication that needs to be sent, the URG pointer is set to 1.
- ACK: The ACK flag is 1 bit and is set to 1 if the acknowledgment number is valid. If the segment does not contain an acknowledgment number, it is set to 0.
- PSH: PSH is the push data flag set to 1 when data needs to be pushed to the application on arrival.
- RST: The RST bit is set when the connection needs to be reset due to a host crash or other reasons.
- SYN: SYN is 1-bit and is used to establish a connection between the sender and the receiver.
- FIN: The FIN flag is set if the connection needs to be terminated.
- Window: This is the size of the window that tells how many bits need to be sent at once when a connection is established.
- Checksum: Checksum is used to detect errors during transmission.
- Options: The options field is variable in length, which is used to add additional features.

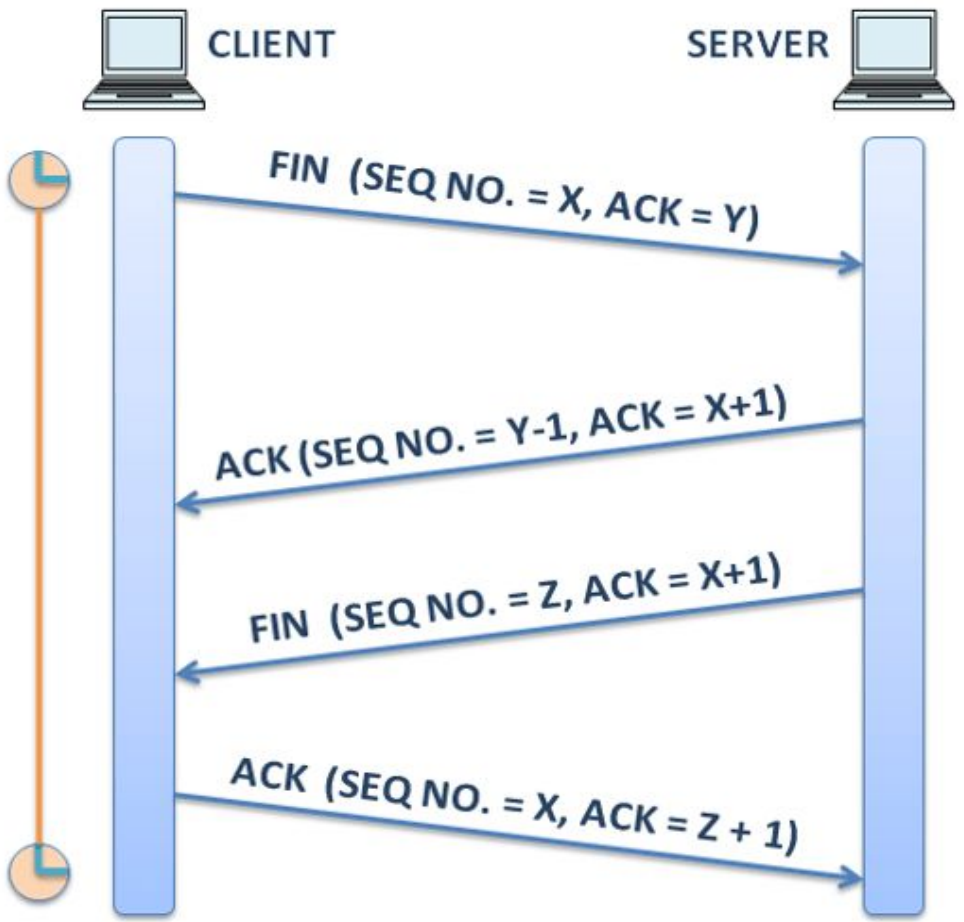
# TCP Connection Establishment

- When two friends meet, they greet each other by shaking hands to start a conversation.
- Similarly, on the network, when the client and the server want to communicate, they use a three-way handshake mechanism.
- The client chooses a sequence number and sends the SYN flag to the server to initiate client-to-server communication.
- When the server receives the SYN flag, it gives an acknowledgment to the client and requests a server-to-client session.
- Here, the ACK is set as the previous sequence number + 1, and the server generates a new sequence number.
- Upon receiving the SYN and ACK fields from the server on the client-side, a connection is established between the client and the server.



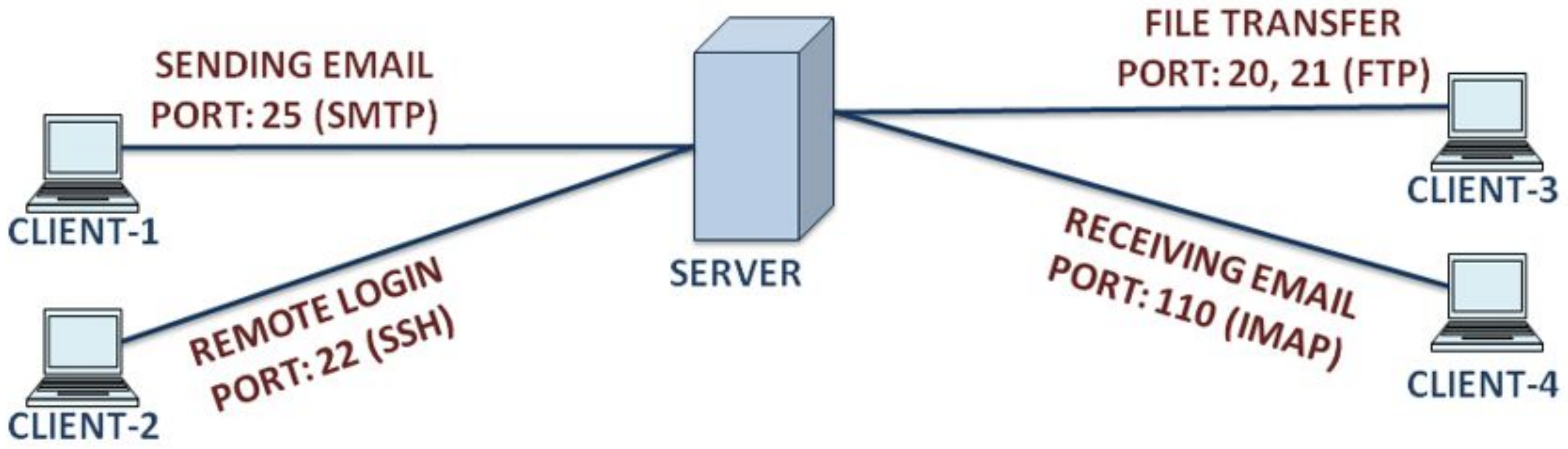
# TCP Session Termination

- When the communication process between the client and the server is completed, the connection needs to be terminated.
- The FIN (FINISH) flag is set in the TCP header to end the session.
- To terminate a TCP connection, there is a two-way handshake mechanism in which the FIN and ACK flags are used.



# Uses of TCP

- When TCP is used, it handles all the tasks associated with the segmentation of data, reliability, tracking and management of data, controlling flow, and reassembling of data.
- For reliable communication, the TCP protocol is used, which guarantees segment delivery.
- If the client does not have a flow and error control mechanism, it can use the TCP protocol because TCP handles flow and error control.



# Socket Programming (TCP & UDP)

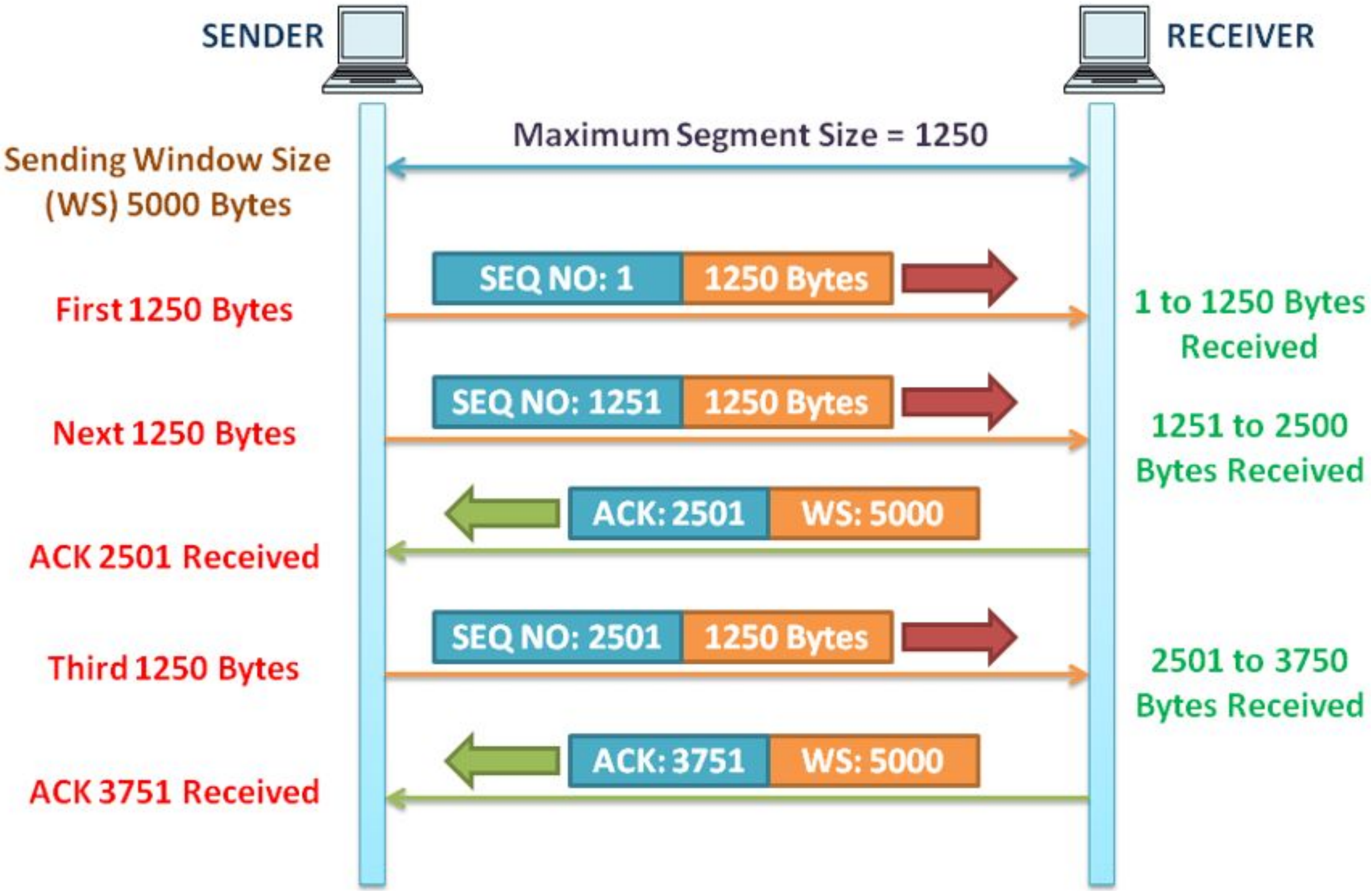
- Program
  - A program is an executable file residing on a disk in a directory.
  - A program is read into memory and is executed by the kernel as ad result of an exec() function.
  - The exec() has six variants, but we only consider the simplest one (exec()) in this course.

- Process
  - An executing instance of a program is called a process.
  - Sometimes, task is used instead of process with the same meaning. UNIX guarantees that every process has a unique identifier called the process ID. The process ID is always a non-negative integer.

- File descriptors
  - File descriptors are normally small non-negative integers that the kernel uses to identify the files being accessed by a particular process.
  - Whenever it opens an existing file or creates a new file, the kernel returns a file descriptor that is used to read or write the file. As we will see in this course, sockets are based on a very similar mechanism (socket descriptors).

# TCP Flow control

- Using a flow control mechanism, the destination receives the amount of data that is reliably processed.
- In short, flow control is used to maintain the reliability of TCP transmissions by adjusting the flow of data.
- To achieve flow control, the window size field of the TCP header is used, which is 16-bit.
- It may be possible that the sender sends a large amount of data. Therefore, there is a limit to how much data the sender can carry before receiving the acknowledgment from the receiver.
- This amount is known as the size of the window, which helps to control the flow.
- The window size is the amount of data that the receiver TCP session can accept and process at once.


SENDER
RECEIVER
Maximum Segment Size = 1250
Sending Window Size (WS) 5000 Bytes
SEQ NO: 1
1250 Bytes
First 1250 Bytes
1 to 1250 Bytes Received
SEQ NO: 1251
1250 Bytes
Next 1250 Bytes
1251 to 2500 Bytes Received
ACK: 2501
WS: 5000
ACK 2501 Received
SEQ NO: 2501
1250 Bytes
Third 1250 Bytes
2501 to 3750 Bytes Received
ACK: 3751
WS: 5000
ACK 3751 Received

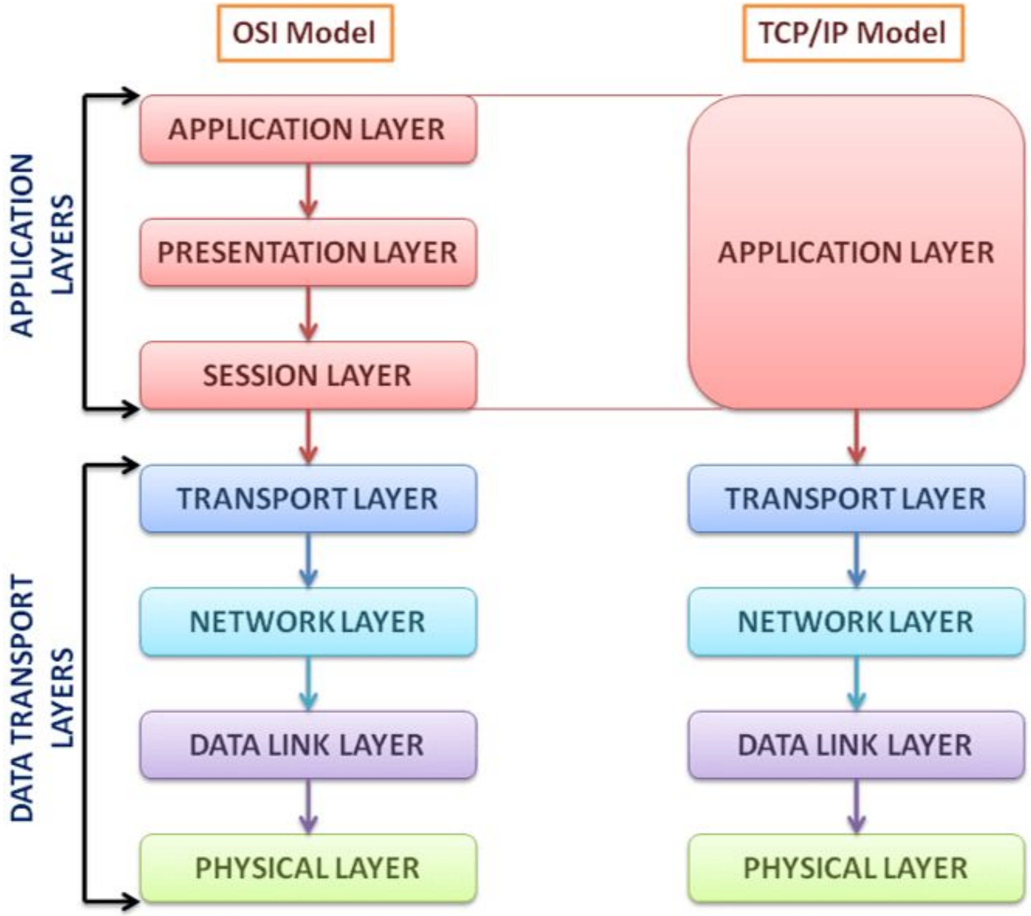
- As shown in the figure, the sender has set the window size for the TCP session as 5000 bytes.
- The window size is present in each TCP segment, so if the window size needs to be modified at any time, the destination buffer can modify the window size based on availability.
- The initial window size is defined during the three-way handshake mechanism.
- First, the sender sends the first 1250 bytes to the receiver, where the sequence number is 1. Immediately after that, the sender again sends the next 1250 bytes to the receiver.
- The receiver receives a total of 2500 bytes, now it sets 2501 as the acknowledgment number and sends it to the sender that it now wants the bytes starting with 2501. It also sends window size as 5000.
- Increases the sender window size to 2500, which becomes 7500. On receiving ACK 2501 from the receiver, the sender sends the next 1250 bytes to the receiver. This process continues until all the bytes from the sender are received to the receiver.
- If the buffer space is reduced, the window size is reduced on the receiver side.
- In today’s networks, the sliding window protocol is used in which the receiver sends an acknowledgment after receiving two segments.
- The main advantage of the sliding window protocol is that it allows the sender to continuously send data, as long as the receiver acknowledges the segments it receives previously.

# TCP Congestion Control

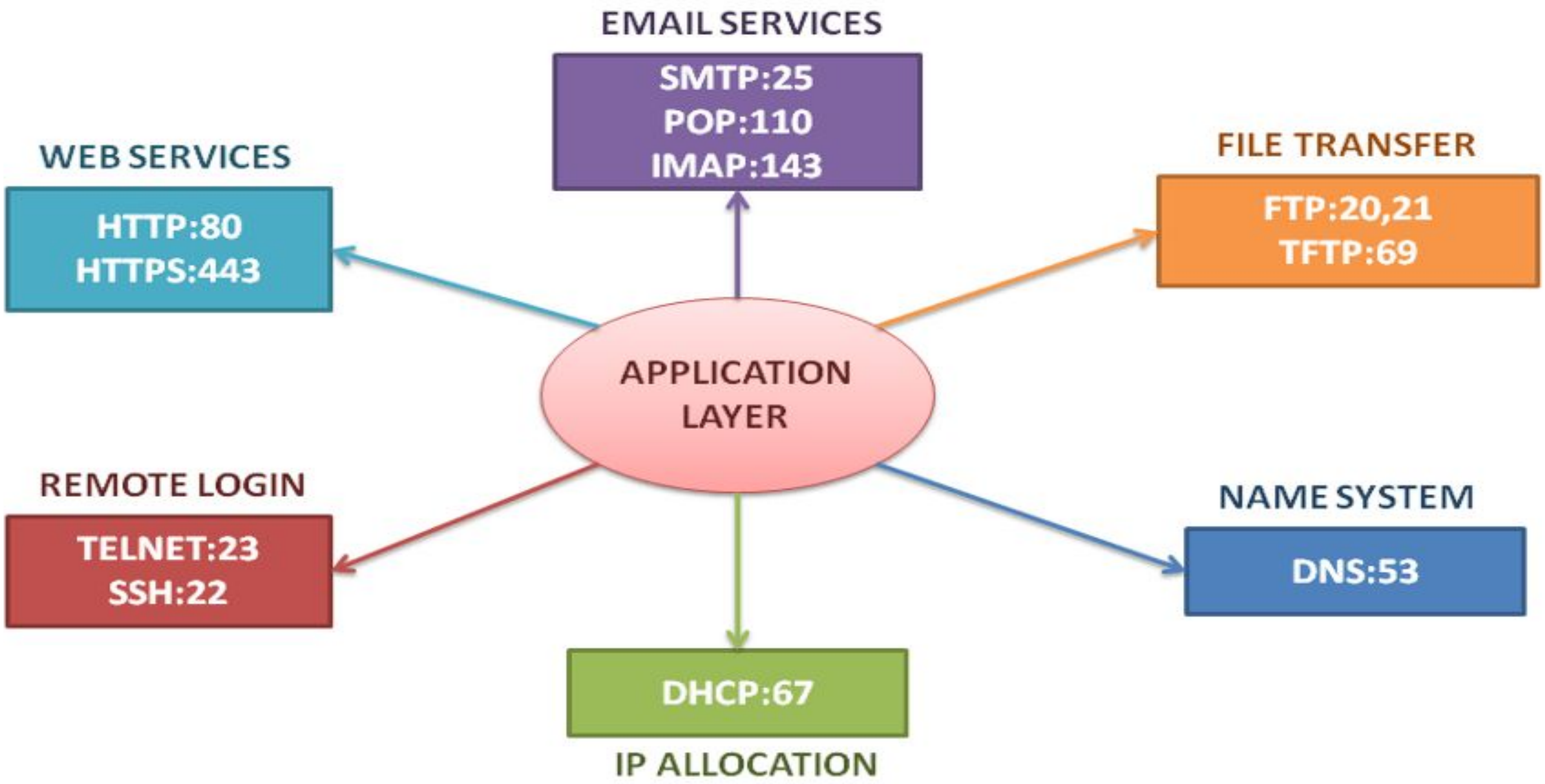
- When network congestion occurs, the packets are discarded by the router and because of this, they are lost in the network. If segments are lost during transmission, they will not reach the receiver and remain unacknowledged.
- The sender learns about network congestion by determining the lost packets and the rate at which TCP segments are transmitted.
- Segments losses are caused by error or traffic. The sender retransmits the lost segments to the receiver.
- But if the retransmission is not done properly by the sender and he is not aware of the congestion, then the existing congestion can get worse.
- The transport layer has mechanisms to avoid congestion. For that, it implements a congestion control algorithm and timer.
- If TCP segment acknowledgment is not received, the sender reduces the flow of data.
- When the sender sends a TCP segment, it starts the timer. If an acknowledgment is received before the timer expires, the sender stops the timer. If the acknowledgment is not received from the receiver, the sender waits for the acknowledgment until the timeout and retransmits the TCP segment.

# Application Layer

- It combines the functionality of the presentation and session layer in the TCP/IP model.
- Generally, it provides an interface to the user whether it is a human or a software program that wants to access the services of the network.
- Services like electronic mail, file transfer, web access, system resource access, and remote login, etc., are provided by the application layer.
- The application layer creates a user-friendly interface to interact with the application program.



# Application Layer Protocols



# Application layer protocols interaction with end-user applications

- 1. Interface Between User and Network
  - End-user applications (like browsers, email apps) do not directly handle networking details. Instead, they rely on application layer protocols.
- 2. Request–Response Communication
  - Most application layer protocols follow a client-server model:
    - Client (User Application) → Sends request
    - Server → Processes and sends response
- 3. Data Formatting and Interpretation
  - Protocols define:
    - Message format
    - Syntax
    - Commands
  - This ensures both client and server understand each other.

## 4. Providing Specific Services to Applications

- Different protocols support different types of applications:

| Protocol | Used By Application | Function |
| --- | --- | --- |
| HTTP / HTTPS | Web browsers | Web page transfer |
| FTP | File transfer apps | Upload/download files |
| SMTP | Email clients | Sending emails |
| POP3 / IMAP | Email clients | Receiving emails |
| DNS | All network apps | Domain name resolution |

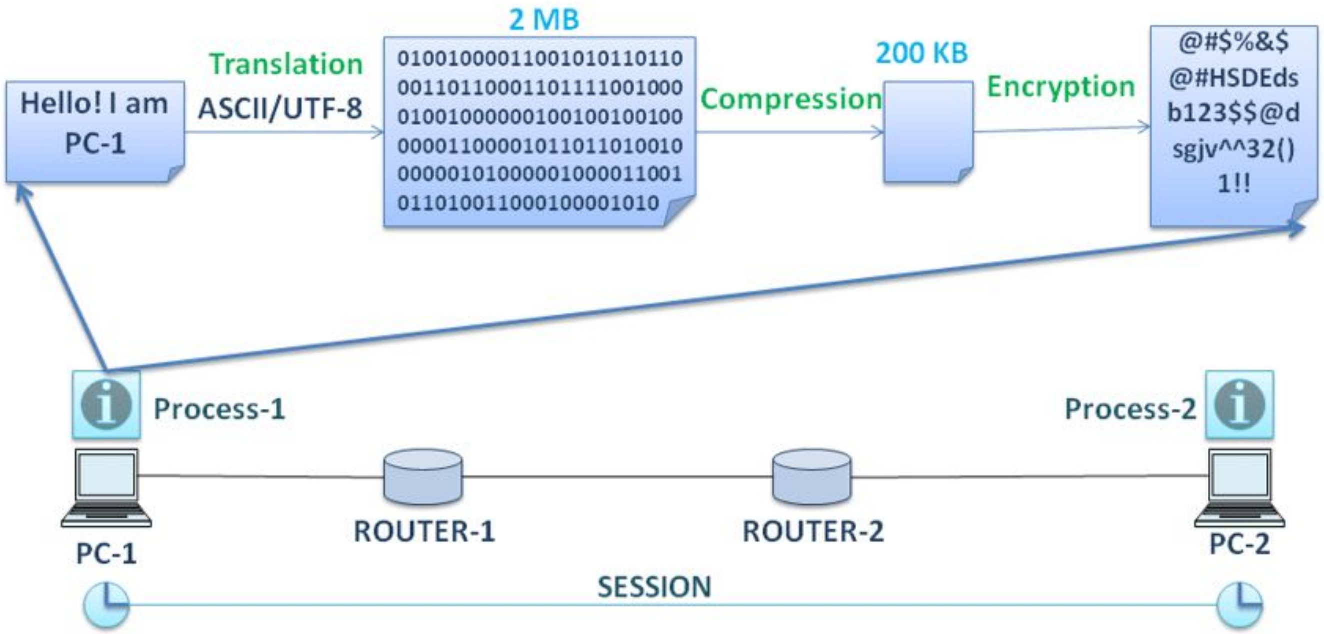
- Interaction Flow (Step-by-Step Example)
  - User enters URL in browser
  - Browser uses DNS → gets IP address
  - Browser sends HTTP request to server
  - Server processes request
  - Server sends response (HTML, CSS, etc.)
  - Browser renders webpage for user

# Presentation layers

- The presentation layer performs translation or formatting, compression, and encryption on the data.
- Translation: When the user gives data to the application layer, the application layer communicates with the presentation layer and the presentation layer translates the data into a format compatible with the receiver.
- Compression: Once the translation is done, the presentation layer compresses the data in such a way that the receiver can decompress it.
- Encryption: After compression, data is encrypted by encryption methods and generates ciphertext so that it can be protected from attackers.
- Basically, the presentation layer contains standards for file formats such as MKV (Matroska Video), MOV (Quicktime Video), Portable Network Graphics (PNG), and the Joint Photographic Experts Group (JPG or JPEG).

# Session layer

- When the user is communicating with the application running at the application layer, the session layer creates a session for the user. It maintains dialog control and synchronization.
- Whatever data the user exchanges in the application layer, the session layer handles all the exchange of information to initiate the dialogs.



# Application Protocols and Services

- HTTP (HyperText Transfer Protocol) and HTTPS (HTTP Secure): For accessing web services.
- FTP (File Transfer Protocol) and TFTP (Trivial FTP): For file transfer.
- DNS (Domain Name System): For conversion of IP to the domain name and vice versa.
- DHCP (Dynamic Host Configuration Protocol): Allocation of IP addresses.
- SMTP (Simple Mail Transfer Protocol): For sending an email.
- POP (Post Office Protocol) and IMAP (Internet Mail Access Protocol): To receive an email.
- Telnet (Teletype Network) and SSH (Secure Shell): Remote Login.

# Web Services

- The application layer provides a user interface to access web services simply with less effort.
- As shown in the figure, the client types the URL of the website, and then a connection is established between the web server and the client.
- Here, the HTTP protocol is used to route the client’s request to the HTML pages. Generally, GET, POST, and PUT messages are used.
- GET, POST, and PUT are used to request a web service from a server, upload data files to a web server, and upload resources to the server, respectively.
- Once the webserver gets the client’s request, it returns the HTML page to the client as a response.
- The main two protocols used for web services are HTTP and HTTPS. Both protocols use a request-response cycle but the data stream is encrypted in HTTPS using the Transport Layer Security (TLS) protocol or the Secure Sockets Layer (SSL) protocol. Hence, HTTPS is more secure than HTTPS.

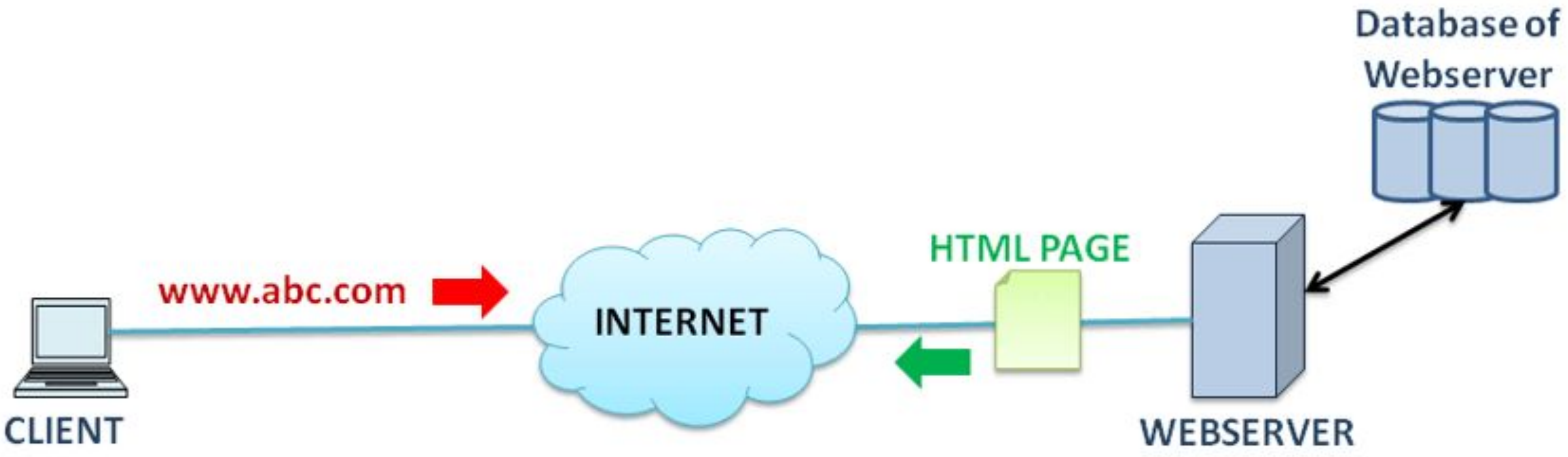



**Figure 5:** Web Services in Application Layer

# Email Services

- Email is one of the primary services of the application layer, and it uses a store-and-forward mechanism.
- When the email is sent by the sender, it is stored in the mail server, and the recipient accesses the email from the mail server. Generally, email service is offered by an Internet Service Provider (ISP).

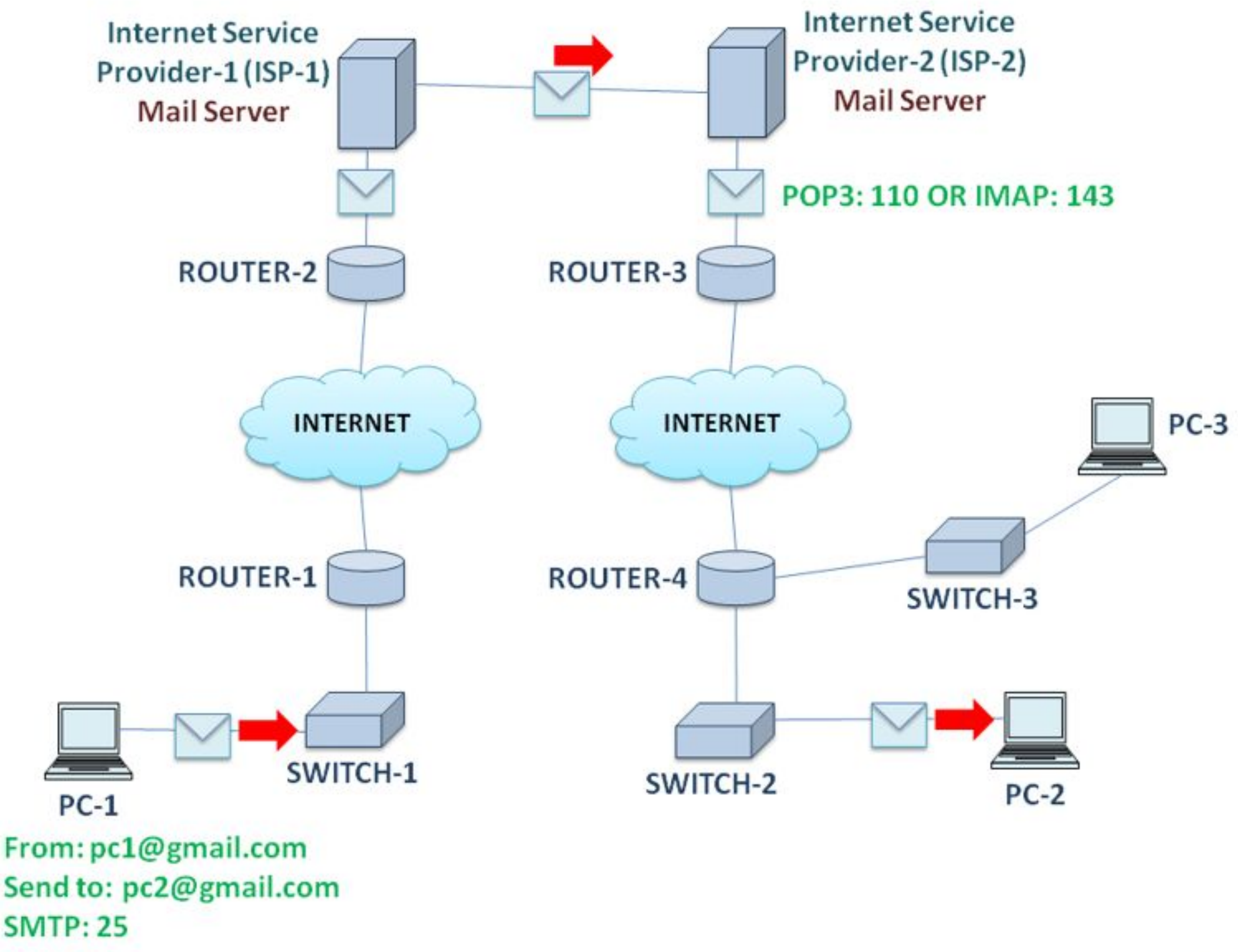



**Figure 6:** Email Services in Application Layer

- As shown in the figure, the sender wants to send an email to the recipient. For that, the sender writes the recipient's email address in the message so that it is received by the appropriate recipient.
- Email messages are protected by the Transport Layer Security (TLS) protocol.
- The application layer has set the SMTP (25) port to send the email. Once an email leaves the sender's application layer, it travels through the router and reaches the mail server.
- The mail server is the intermediate between the sender and the receiver. The mail sent by the sender is not going directly to the recipient. It first reaches the mail server.
- Once the email arrives on the mail server, the recipient uses the POP3 (110) or IMAP (143) protocol to receive the mail from the mail server.
- So we can say that SMTP protocol is used to send an email, and POP3 or IMAP protocol is used to receive email from a mail server.
- Basically, POP is not recommended for small businesses, as they want to back up emails. In this case, they can use IMAP to receive the email because in IMAP, a copy of the original mail is sent to the recipient, and an original email is stored in the mail server.

# IP Addressing Services

- When you connect to a network, an IP address is assigned to you automatically. This is known as dynamic host configuration.
- The DHCP (Dynamic Host Configuration Protocol) server operates on port number 67 and is used to dynamically assign an IP address to the host.
- Also, static IP allocation is used in which the network administrator or engineer has to specify the IP address to host statically.
- DHCP servers are located at different locations on the network. They have an IP address pool in which the configured IP address range is stored.
- The IP assigned to the user is temporary. If the same user connects to a network multiple times, each time he/she gets a unique and different IP address.
- After the communication ends, the IP address is de-allocated.
- The DHCP server sets the time, known as the lease period time. At the end of the lease time, the IP address gets de-allocated and can be reused for any other host.

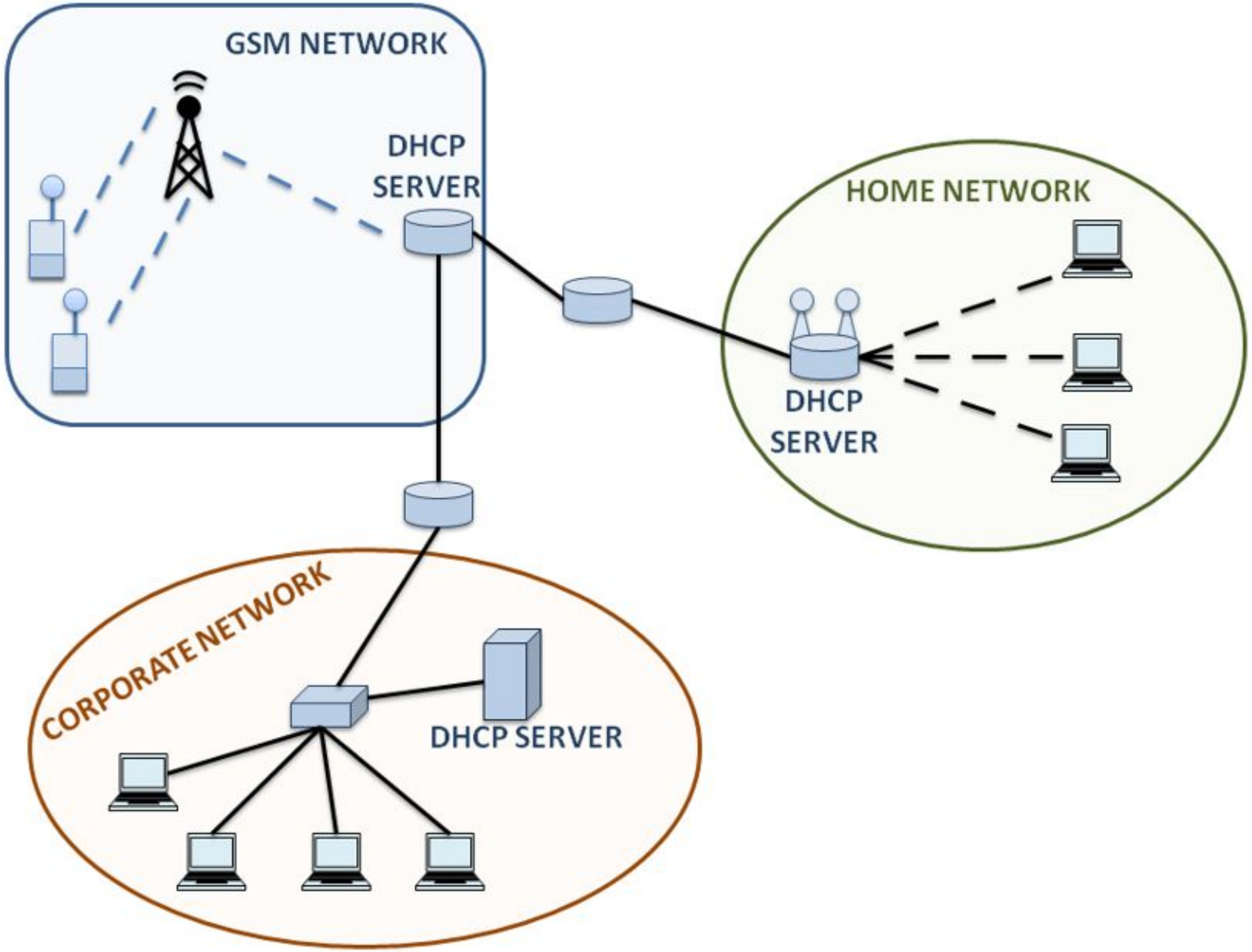
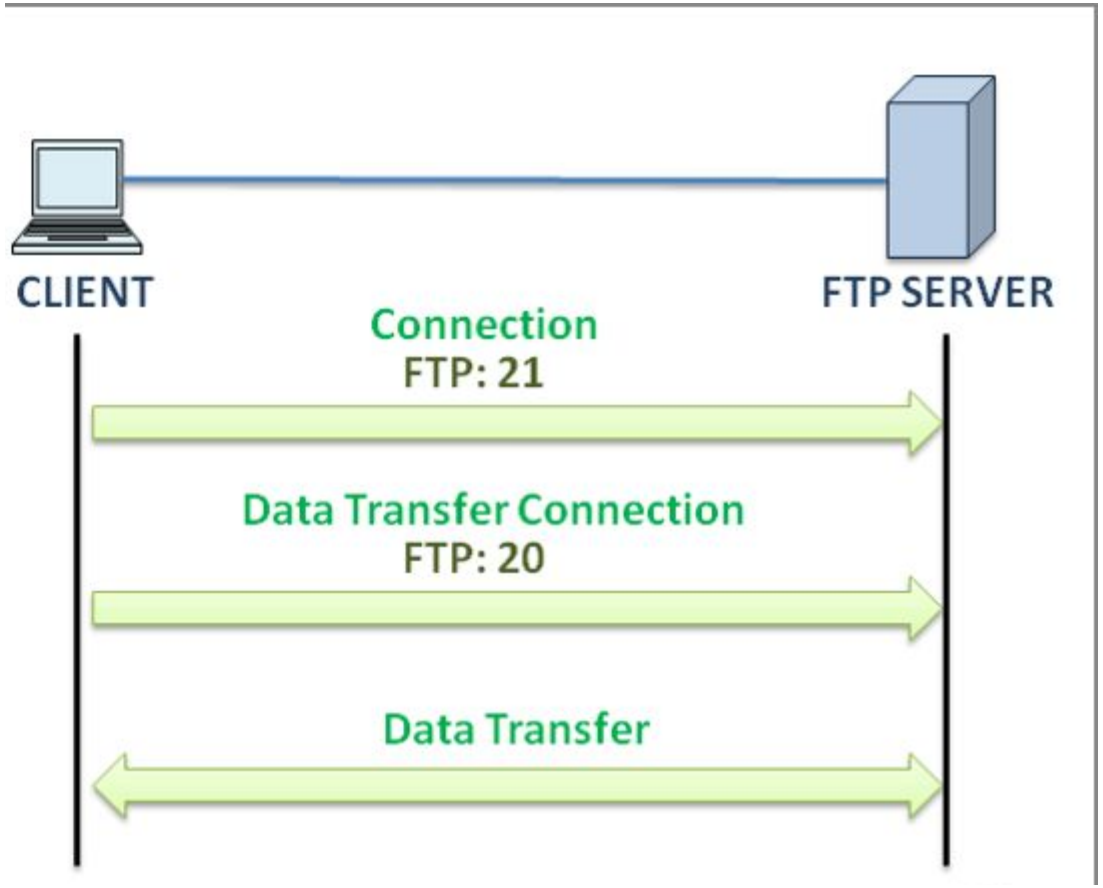



**Figure 7:** IP Addressing Service in Application Layer

# File Transfer Services

- The FTP and TFTP protocols are the primary protocols used at the application layer for file-sharing services.
- FTP is most common application layer protocol used for file sharing between devices.



- As shown in the figure, the client sends a connection request to the server using TCP port 21.
- Once the connection is established, the client sends TCP port 20 for a data transfer connection to the server. It uses TCP port 20 each time the client wants a data transfer service.
- After the data transfer connection is established, the client can download and upload data to and from the server. Downloading data from the server is known as pull, and uploading data to the server is known as a push.
- Server Message Block (SMB) is another file sharing mechanism in which the server makes its resources available to the client.

# Application Layer Services

- **Data Representation**
  - Application layer ensures that data is translated, formatted, compressed, or encrypted appropriately for the receiving application to understand and process.
- **Network Service Access**
  - It enables applications to access network services such as email, file transfers, or remote system access.
  - Email Services: Provides the ability to send, receive, and store email messages over the network.
  - File Transfer Services: Allows the uploading, downloading, and management of files between systems.
  - Web Services: Supports communication between browsers and servers to access web content.
  - Remote Access Services: Enables users to access devices and applications on a remote network.
- **Application Protocols**
  - Application protocols define the rules and methods that enable communication between software applications.
- **Session Management**
  - Application layer establishes, manages, and terminates sessions between communicating applications. It also ensures synchronization and controls the flow of data.

# Domain Name System (DNS)

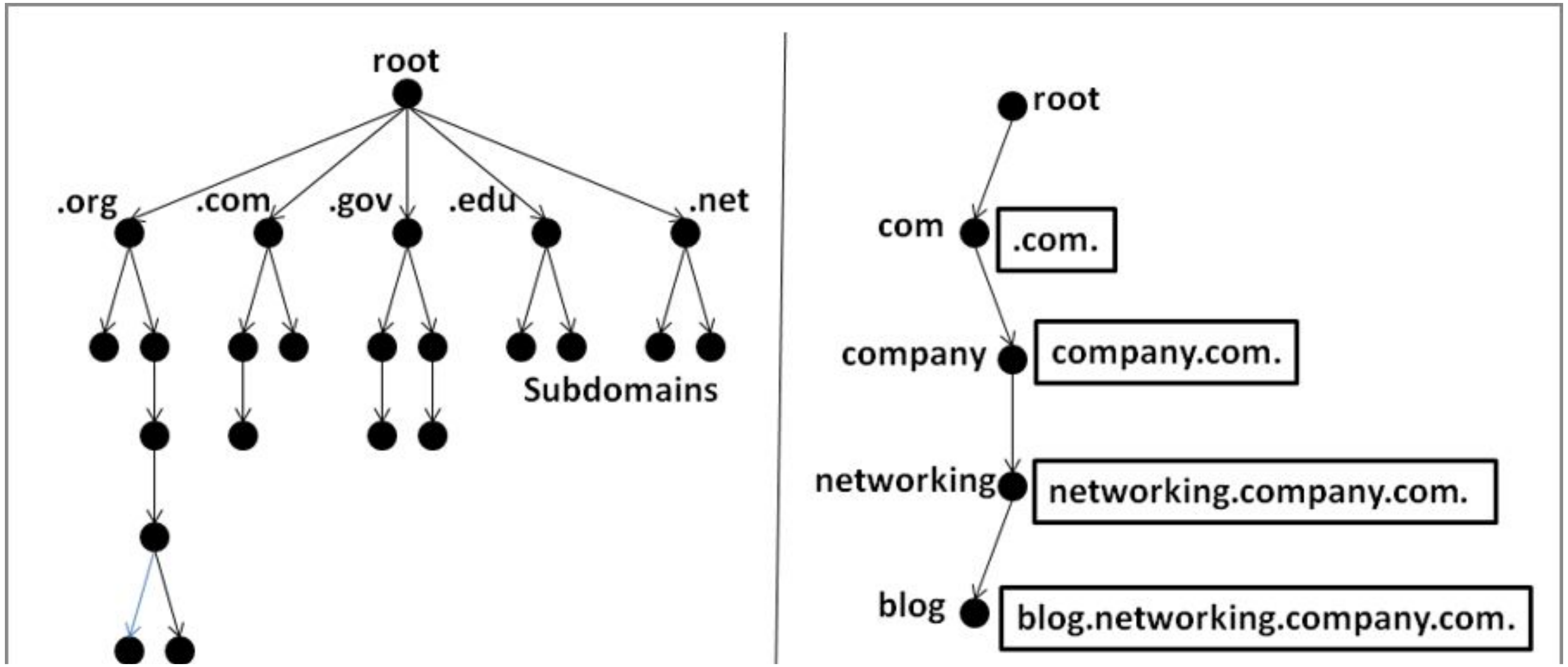
- Domain Name System is also known as Domain Name Service or Domain Name Server.
- Every website has an IP address.
- Every website we access, access through an IP address.
- It takes a lot of time to remember the IP address of a website. That's why DNS is introduced. DNS provides URL because it is easier to remember URL name rather than IP.
- DNS is designed to convert 32-bit numeric IP addresses into simple, easy, and recognizable names.
- For example, a website has the IP address 78.123.45.111, and DNS changes the IP 78.123.45.111 to the simple domain name abc.com. Now remembering abc.com is easier than remembering the 78.123.45.111 IP address.
- DNS includes the format for requests, responses, and data.
- When a user types a website name, the DNS server finds the IP address of that website and displays the result of the query.

# Name Space

- When DNS converts an IP address to a domain name or URL, the URL must be unique.
- No two websites have the same URL.
- Each URL is unique in itself.
- The namespace maps each address to a unique name and can be arranged in a flat or hierarchical format.
  - Flat Name Space: The flat namespace contains the name, which consists of a sequence of characters without structure. For example, in abc.com, abc is referred to as a flat namespace.
  - Hierarchical namespace: In this namespace, the URL is in a hierarchical format, which means that each name is made up of several parts.

# Domain Name Space

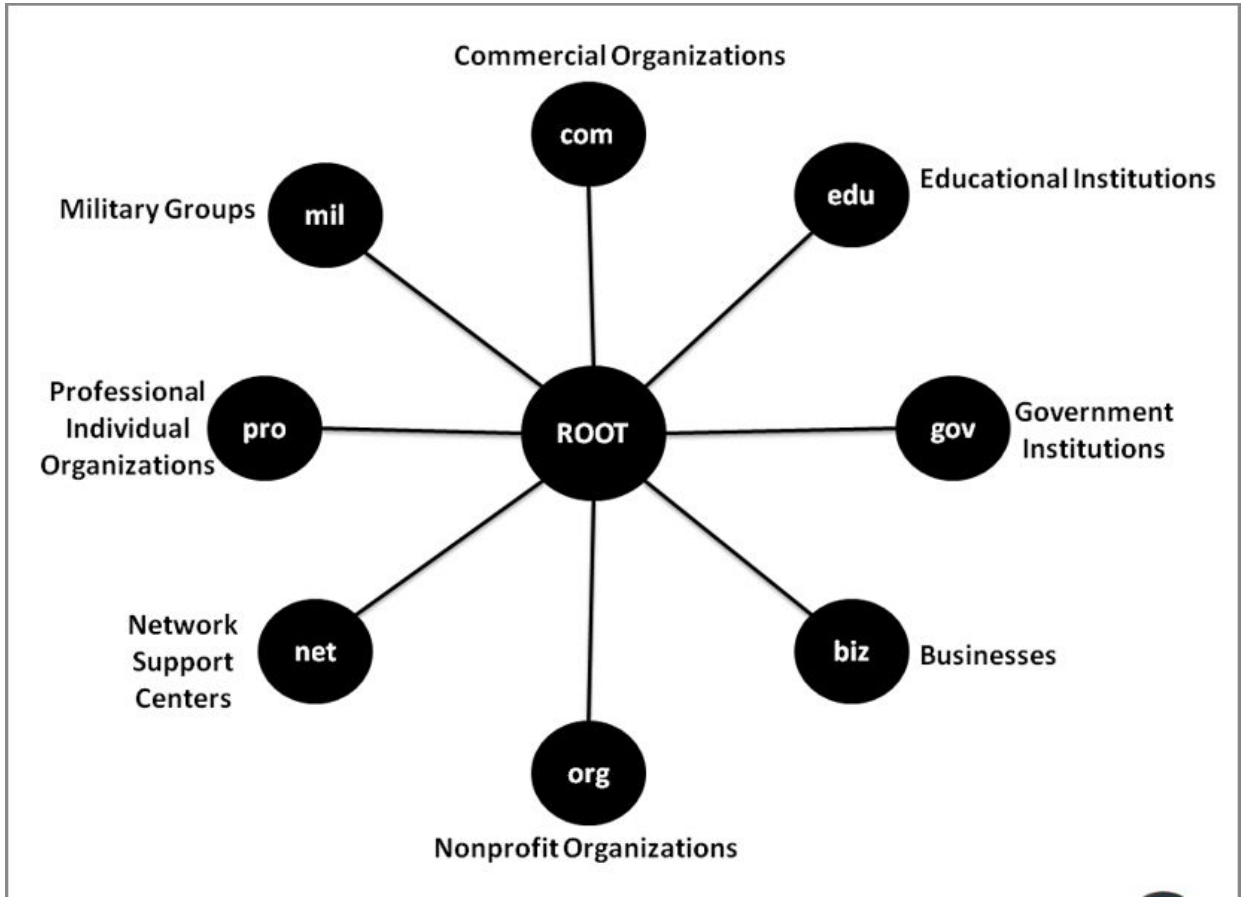
- To have a hierarchical namespace, the domain namespace is used.
- We can define the name as an inverted tree structure with a root.
- The tree has a maximum of 128 levels.



- As shown in the figure, the root node is zero and has children. Also, each node in the tree has a label, a string containing a maximum of 63 characters.
- To achieve the uniqueness of domain names, children nodes of the same parent have different labels.
- Here, the domain name is a sequence of labels, as shown in the figure. For example, blog.networking.company.com. Known as a fully qualified name because a label is terminated by a null string.
- blog.networking.company.com. defines that it is a blog section of the networking section of the commercial company.
- A fully qualified domain name (FQDN) contains the full name of the host, from the most specific to the most general.
- The second is a partially qualified domain name (PQDN), which means that it is not terminated by a null string and does not reach the root. For example, blog is known as a partially qualified domain name because it is not terminated by a null string.

# Types of Domains

- Generic Domain:
  - It defines the behavior of registered hosts. For example, if one wants to open an online business, he can use the .com domain, which defines that the type of business is commercial. Similarly, .org for non-profit organizations, .edu for education, etc.
- Country Domain:
  - Country domain consists of two characters. They use the abbreviated form of the country name. For example, .in for India, .us for USA, .ca for Canada, etc. The address abc.edu.in means ABC college of India and type educational.
- Inverse Domain:
  - Inverse domain is used when mapping an address to a name is required.



# DNS Resolution

- The mapping of names to an IP address is done by the DNS resolution method, this is known as name-address resolution.
  - Whenever a client writes the IP or URL address of the site, the resolver known as the DNS client accepts the query and performs a mapping from the IP address to the name or from the name to the IP address.
  - The DNS client sends the mapping request to the nearest DNS server. If the server has the information, it performs the query operation; otherwise, it refers the resolver to other servers.
  - After receiving the mapping from the DNS server, the resolver also verifies the mapping to check whether it is a real resolution or an error.

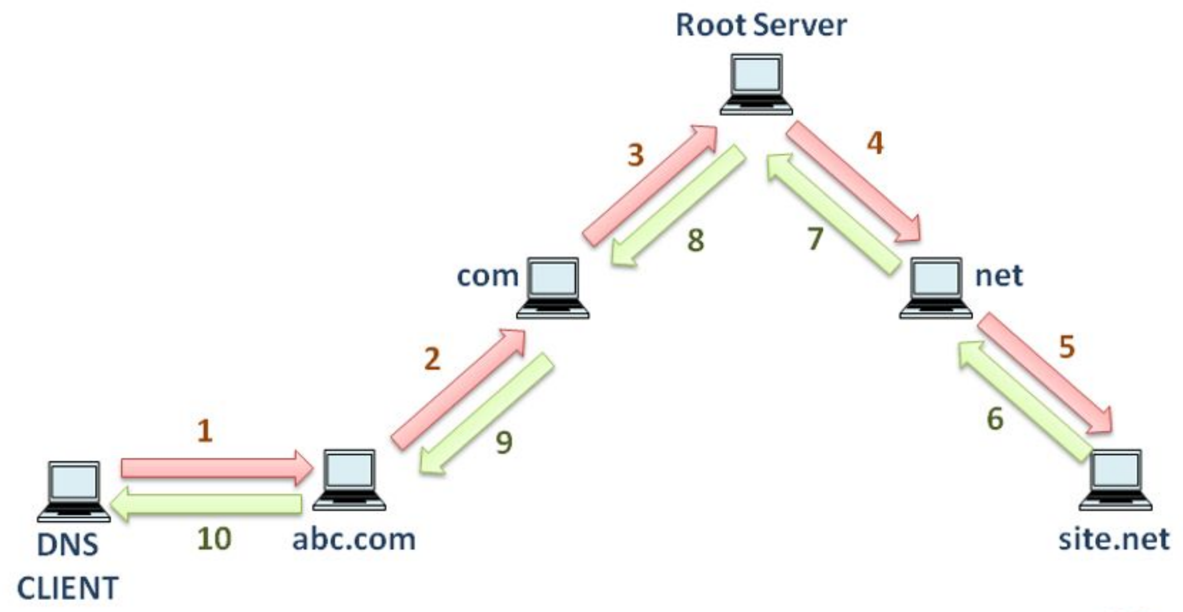



**Figure 5:** Recursive DNS Resolution

# Hypertext Transfer Protocol (HTTP)

- Being a stateless application-layer protocol, HTTP does not retain session information between requests, which limits its ability to handle complex client-server interactions without additional mechanisms like cookies or sessions.
- HyperText Transfer Protocol (HTTP) is a protocol used which transfer hypertext over the Web.
- HTTP has been the most widely used protocol for data transfer over the Web.
- Hyper-text exchanged using HTTP goes as plain text which makes it less secure.
- The web server delivers data to the user in the form of web pages when the user initiates an HTTP request.
- Connection between the web browser and the server ends after the transaction is finished. This makes HTTP a stateless protocol.

- Advantages of HTTP
  - Because fewer connections are running at once, it delivers reduced CPU and memory utilization.
  - It allows requests and answers to be pipelined via HTTP.
  - Because there are fewer TCP connections, it provides less network congestion.
  - During the first stage of connection establishment, handshakes are exchanged. Because there is no handshaking, it provides lower latency for subsequent requests.
  - Without terminating the TCP connection, it reports problems.
- Disadvantages of HTTP
  - It is applicable to point-to-point connections.
  - It isn't mobile-friendly.
  - It sends more data than needed.
  - It doesn't provide trustworthy exchange (in the absence of retry mechanism).
  - When the client receives all the data it requires, the connection is not terminated. Therefore, the server won't be accessible during this time.